

Screen Capture Rectification and Temporal Stabilization for Real-world Captured-screen Videos

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Problem and Motivation

Existing screen demoiréing datasets validate the demand for captured-screen restoration, but their inputs are often controlled, cropped, paired, or spatially/temporally aligned. Real phone-captured screen videos still contain background regions, perspective distortion, hand-held camera shake, weak screen borders, and dynamic screen content. This project targets this missing pre-alignment stage before downstream demoiréing, OCR, or archival restoration.

Task and Goal

Input is a full camera video of a computer screen. Output is a fixed-ratio, front-facing, temporally stable screen video. The goal is to remove most non-screen background, rectify perspective distortion, reduce frame-to-frame jitter, and provide stable screen-coordinate input for downstream video demoiréing, OCR, or archival restoration.

Method

The proposed pipeline first detects or manually specifies the first-frame screen corners, then estimates a homography to rectify the screen to a 16:9 canvas. To avoid frame-wise corner jitter, it tracks reference-plane features with Lucas-Kanade optical flow and estimates a robust homography using RANSAC. Geometric gates reject unreliable updates, and the trajectory is interpolated and smoothed before generating the normalized output video.

Dataset and Experiment Plan

The final evaluation will use a self-collected test set: **5 scenario classes x 10 clips x about 5 s**. The classes are static pages/documents, scrolling webpages, in-screen video playback, PPT or weak-border low-texture pages, and 4K/moiré/glare hard cases. Public demoiréing datasets are used as related-work evidence rather than direct benchmarks for this pre-alignment task.

Evaluation Metrics

We estimate residual affine motion between adjacent normalized frames. Main metrics are residual translation p95, residual rotation p95, and residual scale-delta p95. Tracker accept ratio, RANSAC inliers, inlier ratio, and feature coverage are used to explain failure cases. Lower residual motion means a more stable normalized video.

Initial Result

One local 1920x1080 captured-screen video with 317 frames was used for an initial same-video ablation. The numbers below measure residual motion in the normalized output; they are not ground-truth reconstruction errors.

Method	Last 2 s translation p95	Last 2 s rotation p95	Interpretation
Frame-wise corner detection	1.927 px	0.0425 deg	jittery baseline
Optical-flow tracking	1.929 px	0.0263 deg	lower rotation, still unstable
Reference-plane tracking	0.118 px	0.0044 deg	best residual stability

Expected Result and Timeline

The expected result is a classical geometric preprocessing pipeline that converts realistic captured-screen videos into stable, rectified screen-coordinate videos. Proposal-stage work has fixed the application story, implemented the core normalization pipeline, and produced an initial ablation. Final-stage work will complete the 50-clip test set, run method ablations, visualize corners and tracking failures, and report both successful and unstable cases.

Source files and evidence are stored under deliverables/proposal_20260622/. Current draft uses placeholder Group ID until the official group number is filled.